

Movement Path Estimation and Sewer Condition Analysis

1 Problem Setting

The goal of this project is to estimate the movement trajectory and turning point of an inspection camera traveling inside a pipe, based solely on video frames. Each video consists of a forward motion through the pipe followed by a backward return.

This task is challenging due to several real-world conditions:

- strong visual disturbances such as water, reflections, and sludge,
- varying lighting conditions and exposure levels,
- motion ambiguity in low-texture regions,
- presence of intro/logo frames at the beginning,
- different pipe lengths and recording characteristics.

In addition to estimating the movement path, we implemented a bonus module that analyzes the pipe condition using lightweight computer vision features, producing interpretable outputs and visual summaries.

2 Approach (Movement Estimation)

Our approach combines optical flow-based odometry with a data-driven prior model to ensure robustness across different conditions.

2.1 Optical Flow Odometry

We compute motion between consecutive frames using *sparse Lucas-Kanade optical flow*. Feature points are tracked across frames, and their displacement is used to estimate forward or backward motion.

The pipeline is as follows:

1. Detect the tunnel entry frame to skip intro/logo segments.
2. Select valid frames using quality metrics.
3. Compute frame-to-frame motion using optical flow.
4. Integrate motion signals to obtain a position curve.

The resulting trajectory is scaled to match the known physical channel length.

2.2 Turning Point Detection

The turning point (where motion reverses) is estimated from the position curve using plateau-based detection:

- global maximum (argmax),
- plateau start and end,
- plateau midpoint.

We use the plateau midpoint as the final turning point, as it provides stable results across videos.

2.3 Triangle-Wave Prior

To handle cases where optical flow becomes unreliable (e.g., due to water or blur), we introduce a prior model that assumes a triangular motion profile:

$$x(t) = \begin{cases} \text{increasing (forward),} & t \leq t_{\text{turn}} \\ \text{decreasing (backward),} & t > t_{\text{turn}} \end{cases}$$

The relative turning point is estimated using Leave-One-Video-Out (LOOCV) statistics, ensuring no information leakage from ground truth.

2.4 Ensemble and Smoothing

The final position estimate is computed as a weighted combination:

$$x_{\text{final}} = w_1 \cdot x_{\text{flow}} + w_2 \cdot x_{\text{prior}}$$

Additional refinements include:

- Savitzky-Golay filtering for smoothing,
- monotonicity constraints (strictly increasing before the turning point and decreasing after),
- numerical stabilization.

An optional feature-based regressor is supported but disabled by default due to stability considerations.

3 Bonus: Obstruction and Visibility Detection

For the bonus task, we implemented a lightweight computer vision module to analyze visibility and possible obstructions in the sewer videos. The goal was to generate useful and explainable condition indicators without using heavy machine learning models.

The bonus module provides three main outputs: a per-frame visibility score, a blockage degree, and an obstruction type classification.

3.1 Visibility Score

For each frame, we compute a visibility score between 0 and 1. The score is based on simple grayscale image statistics:

- mean brightness,
- image contrast,
- detection of very dark or overexposed frames.

Frames with moderate brightness and sufficient contrast receive a high visibility score. Frames that are too dark, too bright, or very low in contrast are classified as poor visibility.

3.2 Blockage Degree

To estimate how obstructed or dirty a frame is, we compute the edge density using Canny edge detection. The idea is that debris, rough deposits, roots, gravel, and other obstructions often create many local edges and texture patterns.

The blockage degree is calculated as:

$$\text{blockage} = \frac{\text{number of edge pixels}}{\text{number of image pixels}} \cdot 100$$

This gives a per-frame blockage percentage. Frames above a chosen threshold are grouped into dirty frame ranges.

3.3 Obstruction Type Classification

We also classify possible obstruction types using simple color and texture heuristics. Each frame is converted to HSV color space, and statistics are computed in the center region of the image. These are combined with edge density to classify the frame as one of:

- **Gravel:** medium brightness with high edge density,
- **Lime:** yellowish or whitish deposits with high saturation,
- **Roots:** dark, textured, high-edge structures,
- **Sludge:** dark, low-texture regions,
- **Unknown:** no confident classification.

Consecutive frames with the same predicted obstruction type are grouped into frame ranges. This makes the result easier to interpret and allows the inspection video to be summarized by relevant dirty or obstructed sections.

3.4 Design Choice

The bonus solution is intentionally heuristic and explainable. Since labeled data for obstruction types is limited, we avoided heavy ML models and instead used robust image features such as brightness, contrast, HSV color statistics, edge density, and texture proxies. This makes the output transparent and easy to debug.

4 Conclusion

The combination of optical flow, prior-based modeling, and heuristic condition analysis provides a robust solution for both movement estimation and pipe inspection. While the main model focuses on accurate trajectory reconstruction, the bonus module adds significant value by delivering interpretable insights into pipe conditions.

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